



## CSE422 Lab Project Report

Project Name: Tree Sterility and Survival Analysis

| Group No: 07, Section: 08, FALL 2025 |                     |
|--------------------------------------|---------------------|
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## 1. Introduction:

The project aims to analyze and predict the survival status of trees (represented by the Event variable) based on environmental factors, fungal associations, and soil conditions. Problem Solved: Ecological researchers often need to understand which factors contribute most to seedling mortality. By using machine learning, we can automate the identification of high-risk conditions for tree sterility or death. Motivation: Understanding tree survival is crucial for forest management and reforestation efforts. This project utilizes a real-world ecological dataset to demonstrate the power of classification models and unsupervised clustering in biological research.

## 2. Dataset description:

i. How many features?

Ans: In our data set there are 24 columns which indicates that there are 23 features and 1 target.

ii. Classification or regression problem? Why do you think so?

Ans: This is a Classification problem. The target variable (Event) consists of discrete labels (0 and 1) representing survival status, rather than a continuous range of values.

iii. How many data points?

Ans: The dataset contains 2783 data points

iv. What kind of features are in your dataset? (Quantitative / Categorical)

Ans: In our dataset they have both types of features categorical and quantitative.

```
There are 12 categorical features:
['Subplot', 'Species', 'Light_Cat', 'Soil', 'Adult', 'Sterile', 'Conspecific', 'Myco', 'SoilMyco', 'PlantDate', 'Harvest', 'Alive']
```

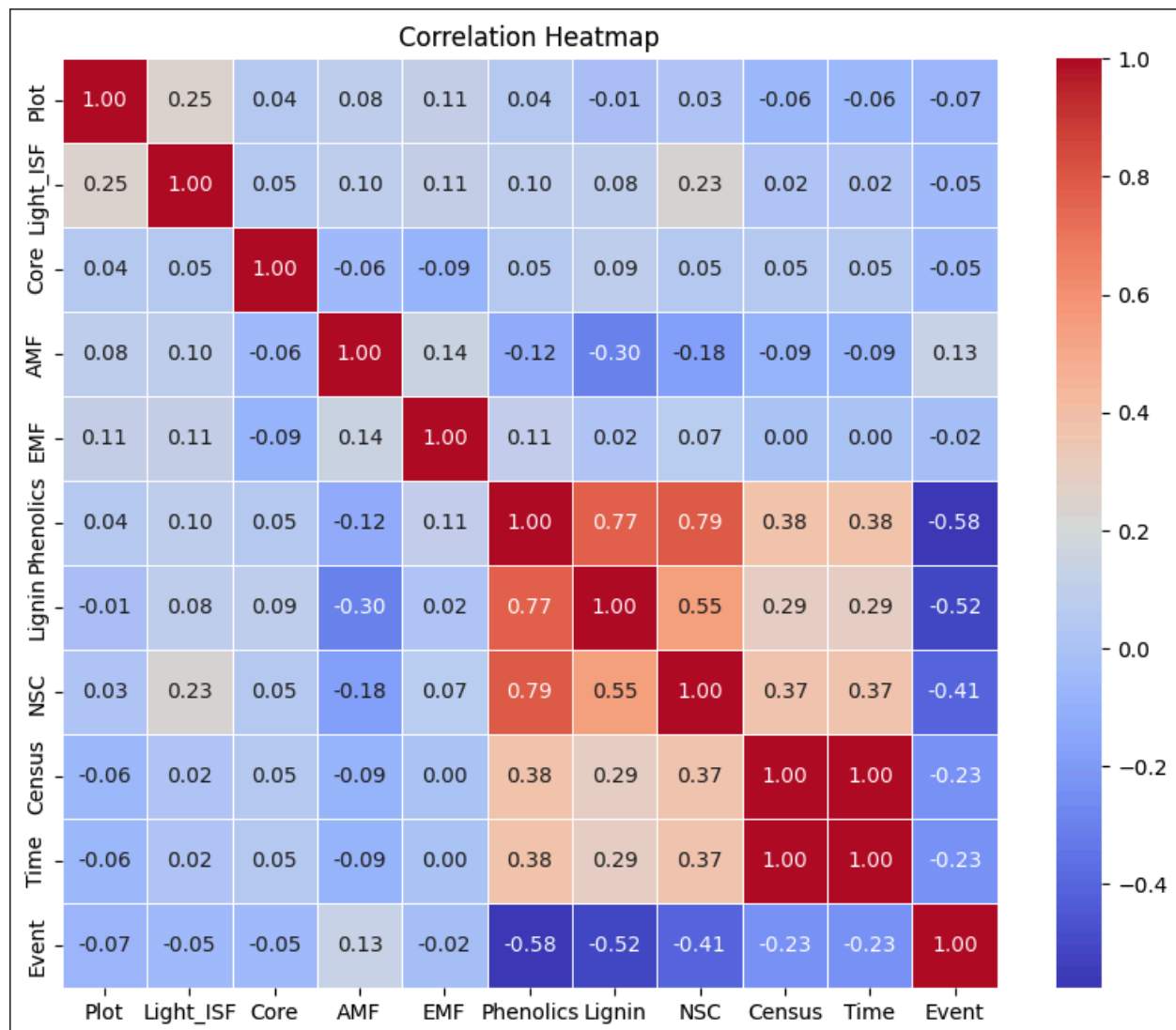
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There are 12 numerical features:
['No', 'Plot', 'Light_ISF', 'Core', 'AMF', 'EMF', 'Phenolics', 'Lignin', 'NSC', 'Census', 'Time', 'Event']
```

v. Do you need to encode the categorical variables, why or why not?

Ans: Yes, categorical features need encoding because ML models require numerical input. Without encoding, the model cannot interpret text-based categories. The choice of encoding method depends on the type of categorical variable.

vi. Correlation of all the features (input and output features) (apply heatmap using the seaborn library)

Ans:



vii. What do you understand after the correlation test?

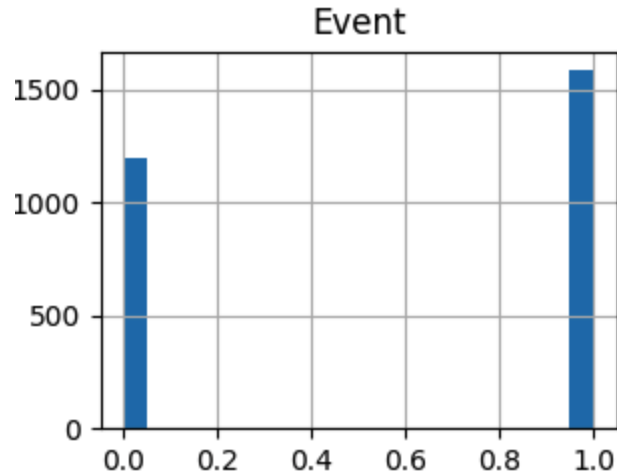
Ans: The correlation heatmap shows that features like Time and Census have a significant relationship with the target Event. Chemical components like Lignin and Phenolics also show moderate correlations with other environmental factors.

### Imbalanced Dataset :

- For the output feature, do all unique classes have an equal number of instances or not?

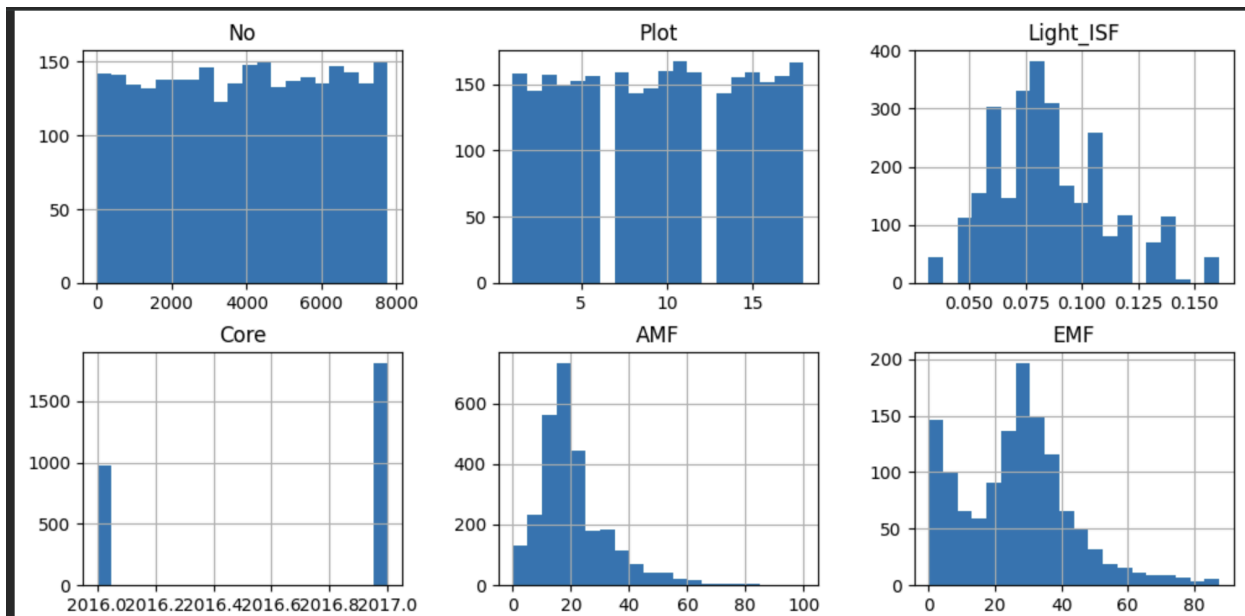
Ans: No, in the output feature all unique classes in the output feature Event are 1.0 (1587 instances) and 0.0 (1195 instances). Here we can see 1.0 has different values and 0.0 has different values. That's why all unique classes don't have an equal number of instances.

- Represent using a bar chart of N classes (N=number of classes you have in your dataset).

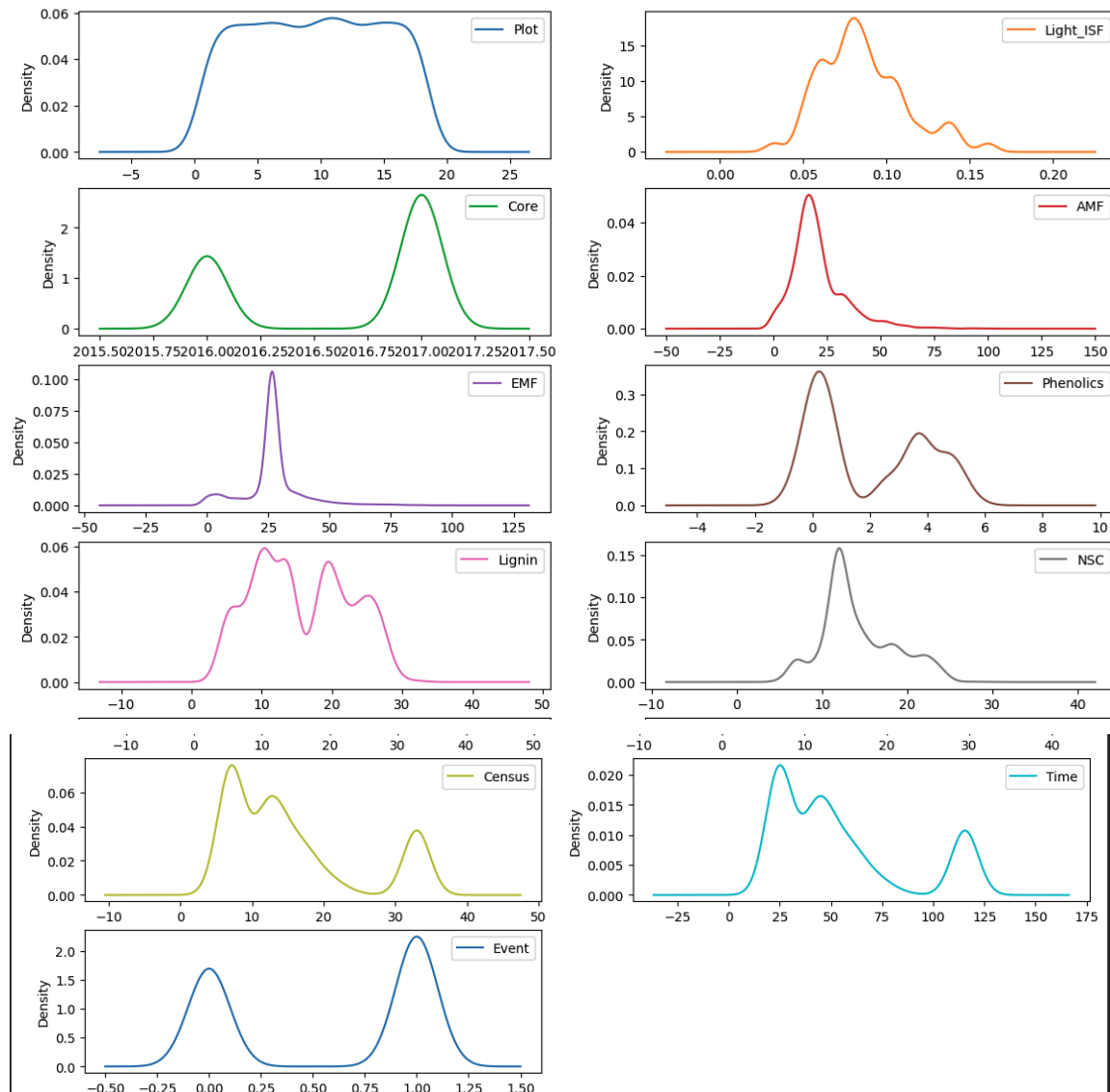


### EDA:

The exploratory data analysis used histograms, density plots, and correlation heatmaps to uncover key patterns in the dataset, most notably that features like Phenolics, Lignin, and Census follow a bimodal distribution rather than a normal one. To ensure data integrity, missing values were filled using the mean, and categorical variables were numerically encoded for model compatibility. Correlation tests (Pearson, Spearman, and Kendall) consistently identified Phenolics and Lignin as having the strongest negative impact on the Event target, providing a clear roadmap for feature selection. Finally, the plots reveal a class imbalance in the target variable, highlighting the need for specific strategies to ensure the predictive model remains accurate and unbiased.







### 3. Dataset pre-processing:

#### • Faults & Solutions:

The dataset needed various pre-processing processes before model training to ensure correctness and consistency.

i. Null / Missing values: in our data set there were so many null values in Harvest (2079), Alive (2292) and also there was some irrelevant column like No, plantDate, adult so for our future calculations we dropped them. Also we replace NaN value with mean value in the EMF column.

ii. Categorical values: Machine learning models work with numbers, not text. So, categorical columns (like “Subplot”, “Species”, “Light\_cat”, “soil”, “Sterlie”, “conspecific”, “Myco”, “SoilMyco”) must be converted into numbers for the model to understand. We used One-Hot Encoding because the categories have no hierarchy (no order or ranking), so each category is represented as a separate binary column.

iii. We used Standard Scaler to scale features so that they have a mean of 0 and standard deviation of 1, helping the model learn efficiently.

#### **4. Dataset splitting:**

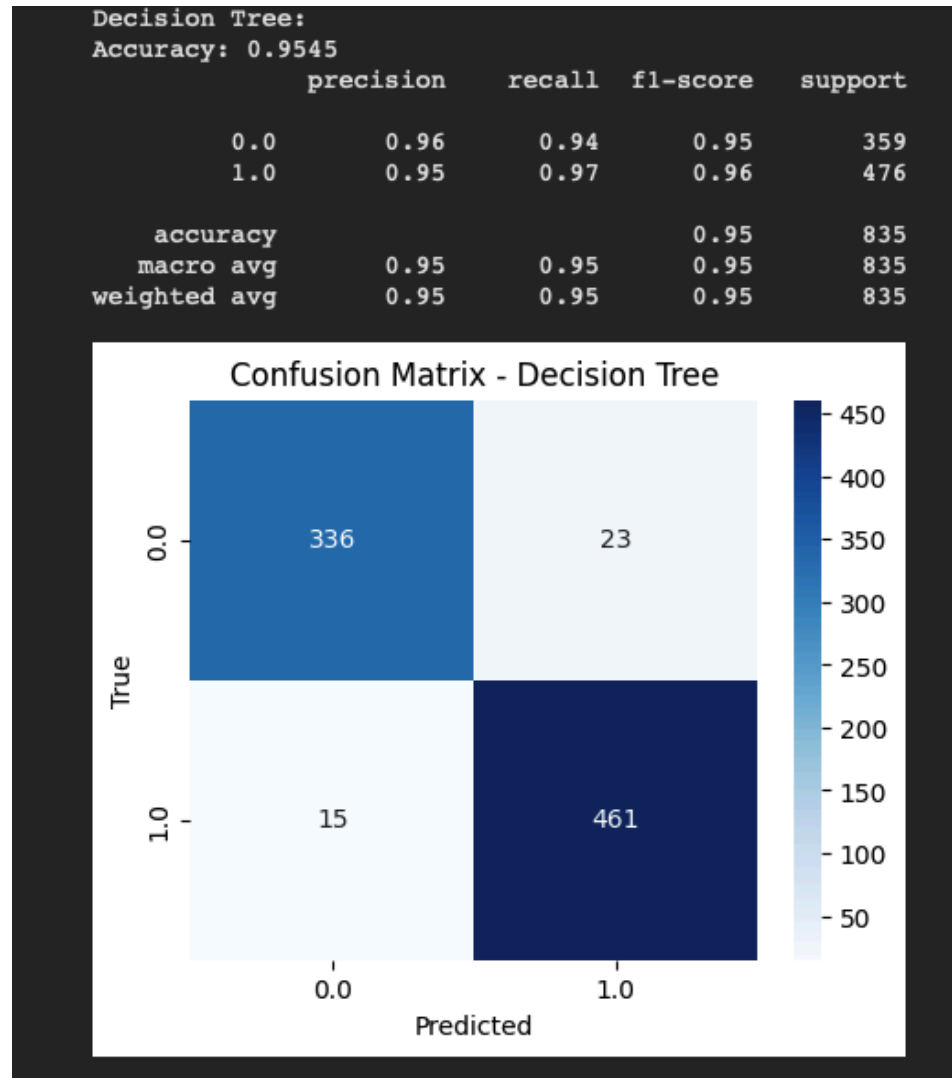
The dataset was separated into training and testing subsets after pre-processing. A 70-30 split was used, with 70% of the Training Set used for model training and testing Set is 30%, for Model performance evaluation. Random sampling was used to separate the target variable without stratification.

#### **5. Model training & testing (Supervised)**

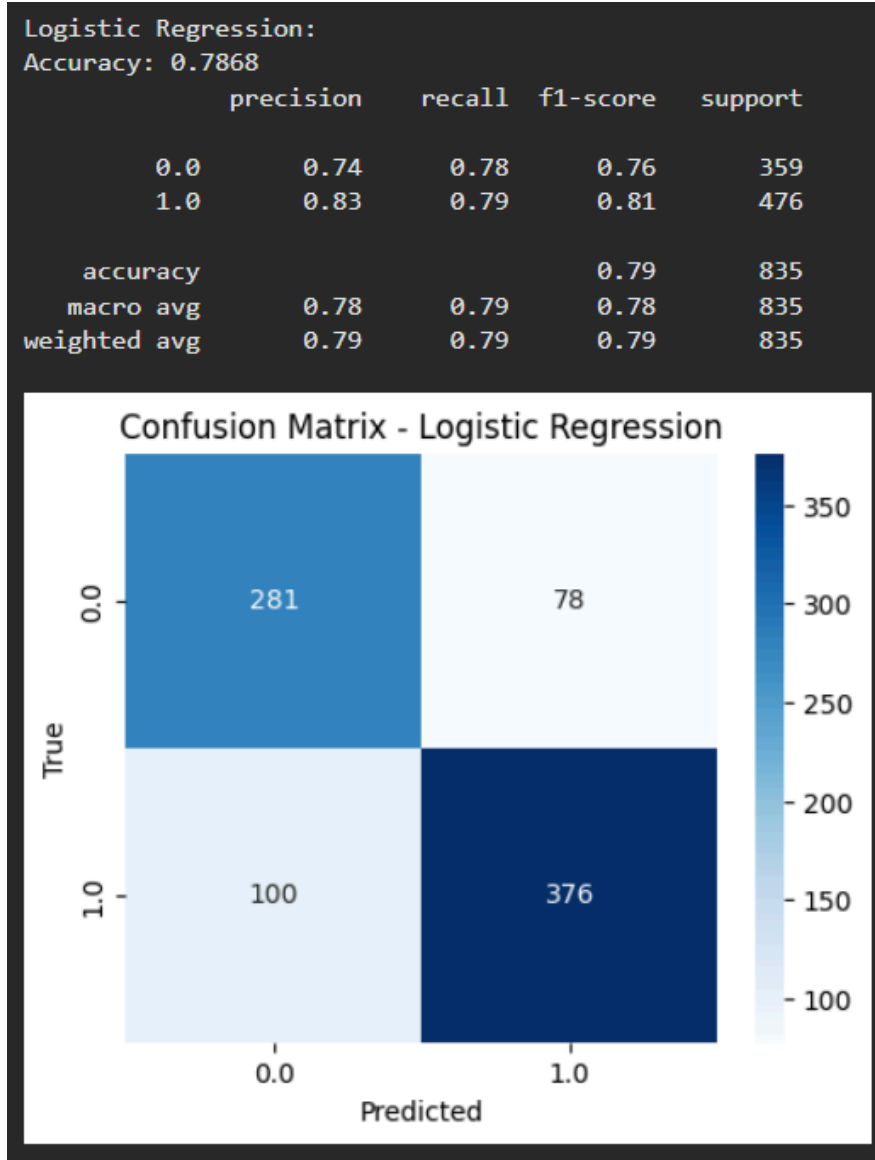
Used models: Decision tree, Logistic regression and Neural Network

- Decision tree: Decision Tree model was trained, which splits data based on feature thresholds to form a tree-like structure. This model is easy to interpret and handles both categorical and numerical data. It can automatically capture non-linear relationships between features and the target variable.

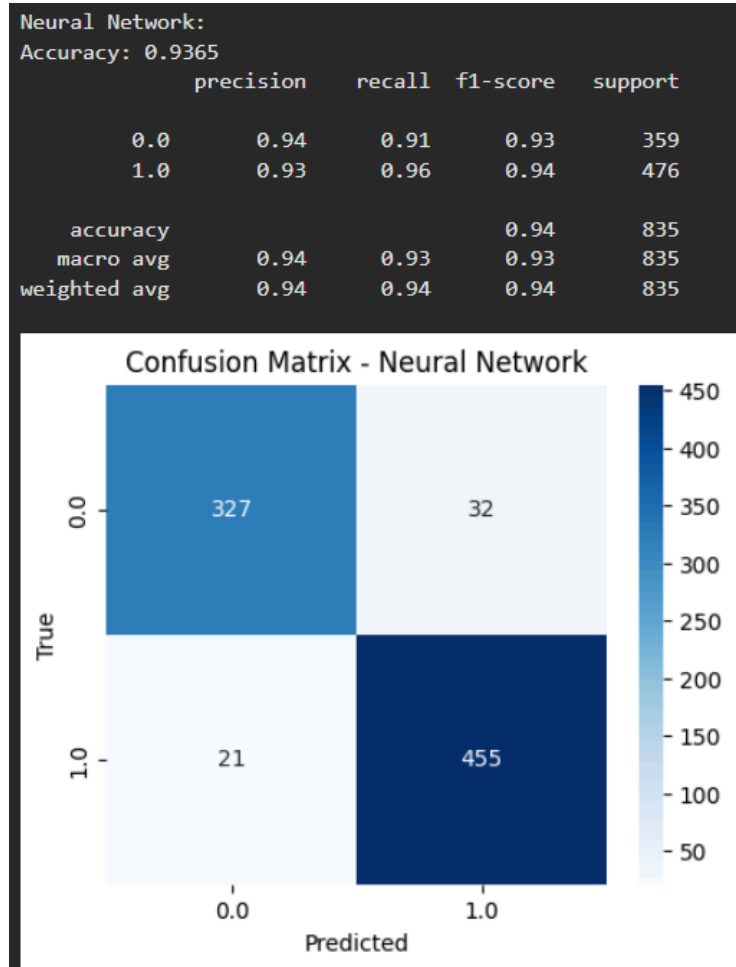




- Logistic regression**: The target variable event has 2 classes (0 and 1), making this a binary classification problem. Logistic Regression can model the probability of the event occurring. This model provides coefficients that show the relationship between features and the target variable, making it easier to understand what influences the event.

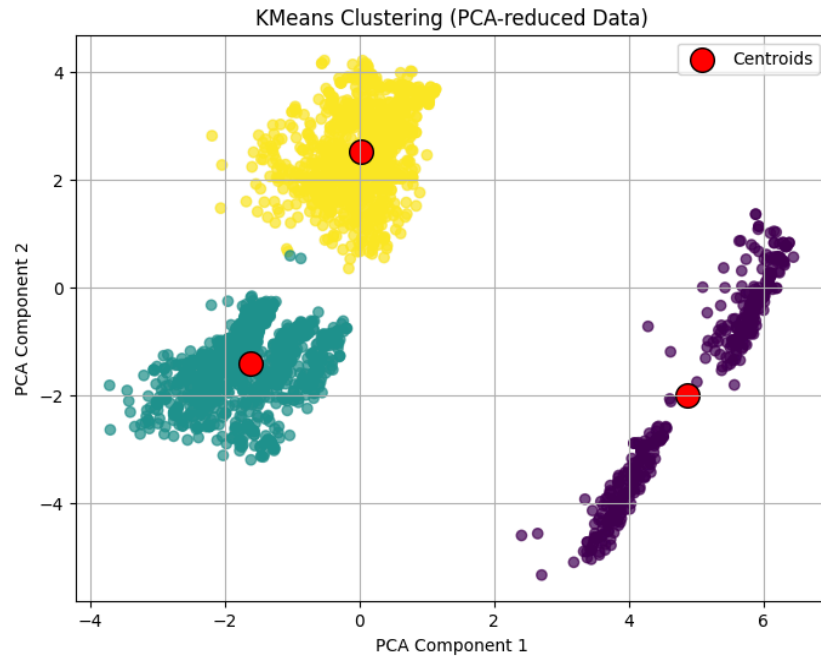


- Neural network: This model automatically discovers relevant patterns and feature interactions without manual feature engineering. For complex problems like tree sterility and survival analysis, neural networks often achieve better predictive performance.

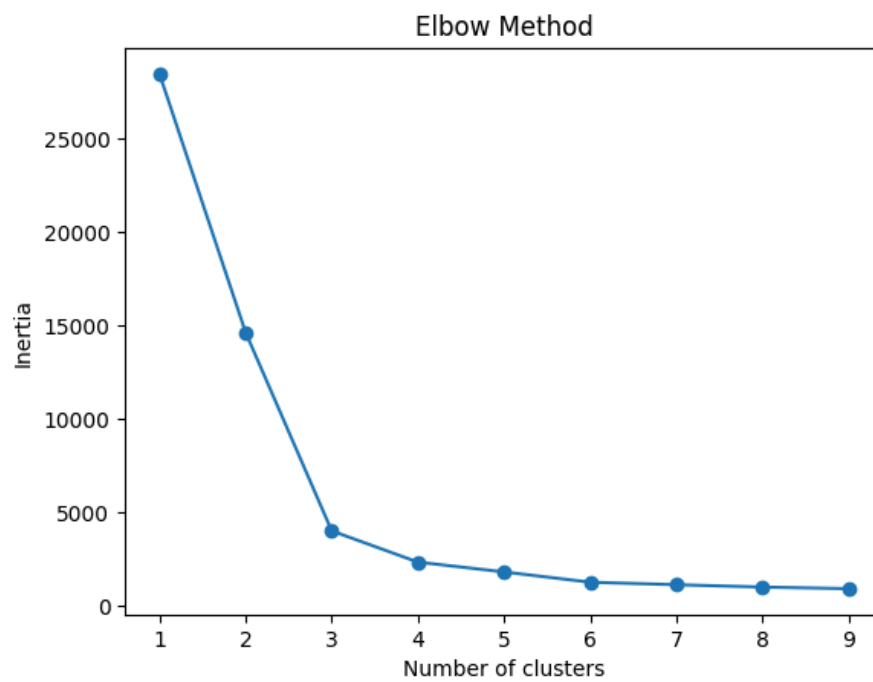


- Unsupervised: K Means:

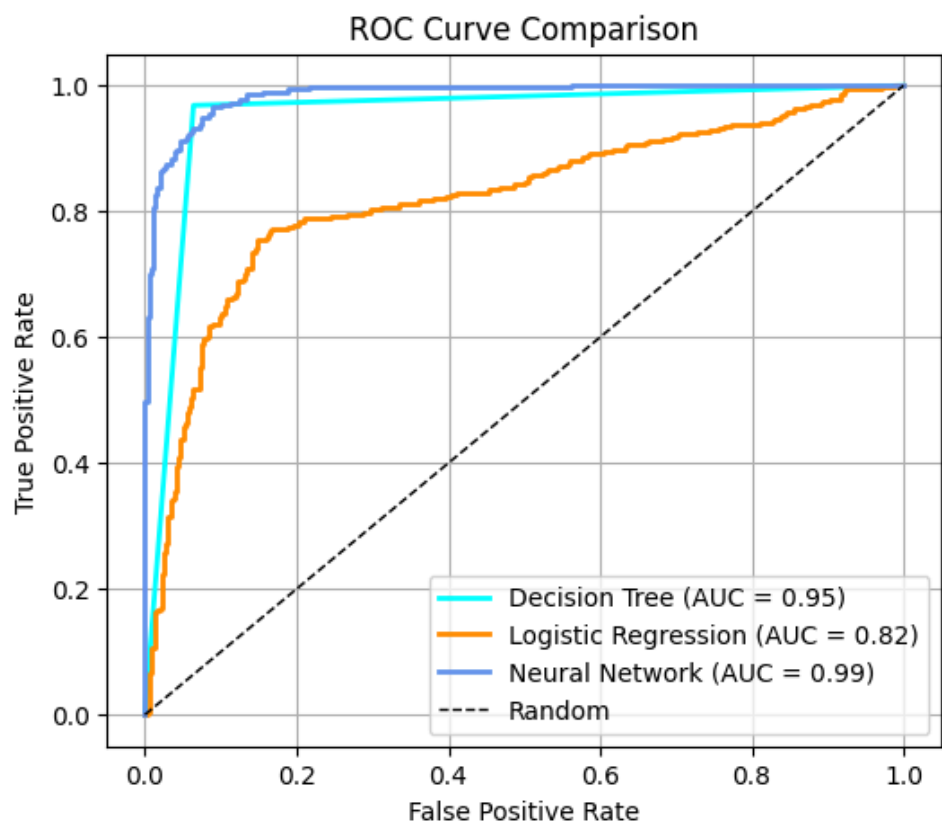
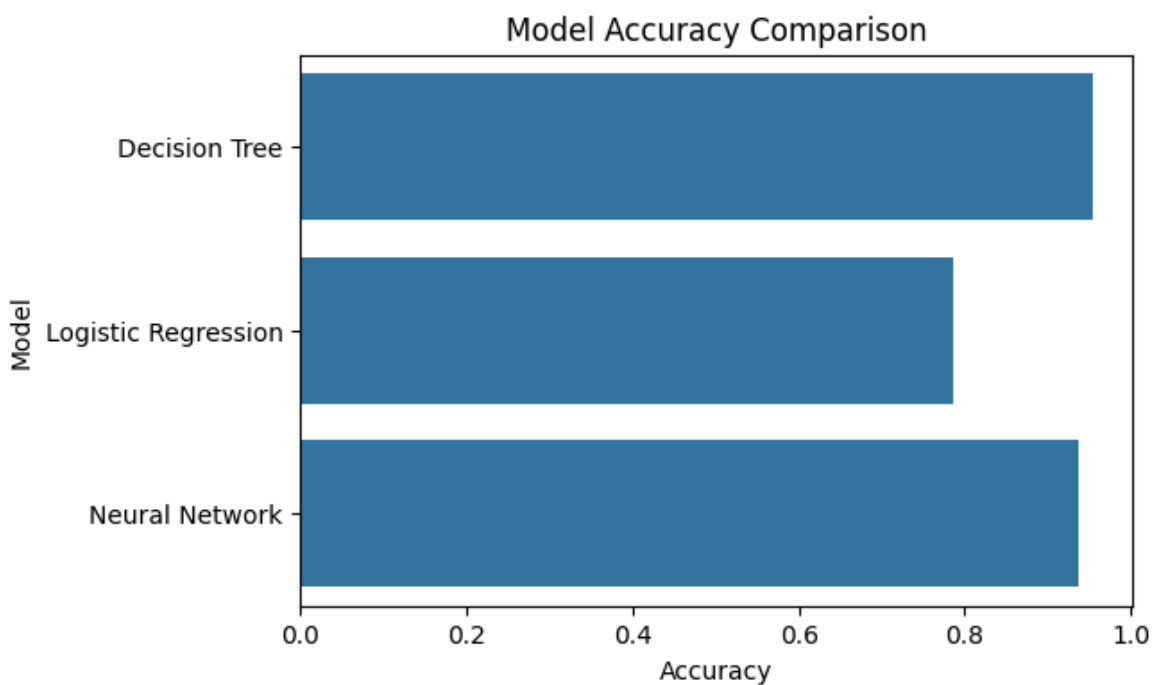
A  $k = 3$  KNN classifier was used. This algorithm classified data by the closest neighbor majority class.



- Elbow method:



## 6. Model selection/Comparison analysis:



While the Decision Tree has the highest accuracy (0.95), the Neural Network achieves the highest ROC-AUC (0.99), indicating it is better at distinguishing between the classes. Logistic Regression has lower accuracy (0.79) and ROC-AUC (0.82), suggesting it captures fewer complex patterns. Overall, models like Neural Network and Decision Tree are more suitable for this problem due to their ability to handle non-linear relationships in the data.

## **7. Conclusion**

Tree survival can be predicted effectively by combining chemical features with environmental variables. Among the models, the Decision Tree achieved the highest accuracy (95%), followed closely by the Neural Network (94%), while Logistic Regression performed lower (79%), suggesting simpler models may miss complex patterns. Some missing data in key features posed a challenge, and imputing values was necessary, though it may have introduced slight noise. Overall, Decision Tree and Neural Network proved most effective for predicting tree sterility and survival.